

AdForge AI: Project Documentation

Date: 1 July 2026

Developed by: Rudra Shekhawat

Project Overview: AdForge AI is an advanced Advertisement Copy Generator utilizing Groq API, Flask, and secure deployment via Ngrok.

1. Dashboard Overview

The AdForge AI project is managed via the SkillWallet portal, which structures the development journey into discrete, manageable epics.

Project Details

Project Name: AdForge AI: AI-Powered Advertisement Copy Generator

Assigned to: Rudra Shekhawat

Status: Under Review

2. Technical Prerequisites & Skill Bank

To execute this project, the following technical competencies are required and linked within the dashboard:

- **Flask Framework:** Essential for backend routing and server-side logic.
- **Groq API:** The core inference engine for AI text generation.
- **Web Fundamentals:** HTML, CSS, and JavaScript for the responsive UI.
- **Version Control:** Git integration for code management.
- **Deployment:** Ngrok configuration for public-facing URLs.

3. Epics Roadmap

The project follows a structured progression:

- **Epic 1:** Model Selection and Architecture.
- **Epic 2:** Core Functionalities Development.
- **Epic 3:** Backend Development.
- **Epic 4:** Frontend Development.

4. Implementation: Backend Logic

The Flask application serves as the main controller. It handles requests, interacts with the Groq client, and returns JSON-formatted advertisement copy.

5. Implementation: Frontend UI

The UI is designed for professional utility, utilizing glassmorphism principles to provide an intuitive user experience for copy generation.

6. Implementation: AI Integration

Utilizing Llama 3 models via Groq SDK ensures low-latency generation of highly persuasive advertising copy.

7. Security Best Practices

API keys are never hardcoded; they are managed through local environment files (.env) to prevent leaks.

8. Deployment Pipeline

Using Ngrok allows for public exposure of the local development server, enabling real-time testing and demo capabilities.

9. Testing Protocols

Systematic verification of all endpoints ensures robustness across different advertisement topics and tones.

10. Performance Benchmarking

Monitoring request latency and throughput is critical for maintaining a high-quality user experience.

11. Future Scope: Integration

Future plans include connecting with social media APIs for direct posting capabilities.

12. Future Scope: Analytics

Implementing a feedback loop based on engagement metrics to fine-tune copy generation quality.

13. Documentation Strategy

Maintaining modular code and clean API documentation allows for future scalability and maintenance.

14. Project Summary

AdForge AI represents a significant achievement in automating marketing workflows through LLM integration.

15. Conclusion & Contact

Final sign-off on technical requirements and project completion pending final review.